

DP-900 Azure Data Fundamentals

Complete cheat sheet · Exam-ready reference

① Core Concepts ~25% ② Relational Data ~25% ③ Non-Relational Data ~20% ④ Analytics Workloads ~30%

1 Core Data Concepts~25%

🔍 5 Analytics Types

Type	Question answered	Example
Descriptive	What happened?	Monthly sales dashboard, KPI report
Diagnostic	Why did it happen?	Why did Q3 sales drop?
Predictive	What will happen?	Will this customer churn?
Prescriptive	What should we do?	Offer 10% discount to prevent churn
Cognitive	Perceives unstructured data via AI	Closed captions, speech-to-text, image tagging, translation

⚠ **TRAP:** "Creating closed captions from audio" → **Cognitive**, NOT descriptive. Cognitive = AI perceiving audio/video/images.

Order: Descriptive → Diagnostic → Predictive → Prescriptive → Cognitive (past → future → action → AI)

📁 Data Types & File Formats

Type	Structure	Examples
Structured	Fixed schema, rows & columns	SQL tables, Excel
Semi-structured	Schema flexible, self-describing	JSON, XML, YAML, Avro
Unstructured	No schema	Images, video, audio, PDF

Format	Type	Best for
CSV	Row-based text	Simple data exchange
JSON	Semi-structured	APIs, document stores
Parquet	Columnar binary	Analytics, Synapse, ADLS
Avro	Row-based binary	Streaming, serialization
ORC	Columnar binary	Hive/big data workloads

👥 Data Job Roles

Role	Core responsibility	Exam trigger words
Data Engineer	Build pipelines, move & transform data, manage infrastructure	"pipeline", "transfer data", "data warehouse ingestion"
Database Administrator	Create & manage databases, security, backups, performance	"create a transactional database", "manage access"
Data Analyst	Query data, build reports & visualisations, surface insights	"chart", "report", "dashboard", "visualise"

Data Architect

Design overall data strategy and system architecture "design the architecture", "overall strategy"

⚡ OLTP vs OLAP

	OLTP (Transactional)	OLAP (Analytical)
Purpose	Day-to-day transactions	Analysis, reporting
Operations	INSERT, UPDATE, DELETE	SELECT, aggregations
Data volume	Current, small per query	Historical, huge datasets
Schema	Normalised (3NF)	Star/snowflake schema
Azure service	Azure SQL Database	Azure Synapse Analytics
Response time	Milliseconds	Seconds to minutes

⬆️ IaaS vs PaaS vs SaaS

Model	You manage	Microsoft manages	Azure example
IaaS	OS, engine, backups, patches	Hardware, networking	SQL Server on Azure VM
PaaS	Data, access config	OS, engine, backups, patches	Azure SQL Database, Cosmos DB
SaaS	Your own data/content	Everything	Microsoft 365, Power BI (service)

🔗 **PaaS facts for exam:** No OS patching · Automatic backups · You do NOT need to apply database engine upgrades · Microsoft 365 subscription NOT required for Azure services.

2 Relational Data on Azure

~25%

📊 Azure SQL Service Comparison

Service	Model	SQL Server compat.	Management	Pick when...
Azure SQL Database	PaaS	High (some limits)	Fully managed	New cloud-native app, minimal features needed
Azure SQL Managed Instance	PaaS	~99% (near full)	Fully managed	Lift-and-shift + needs SQL Agent, Database Mail, Service Broker, linked servers
SQL Server on Azure VM	IaaS	100%	You manage	Full control needed, legacy features, OS access required

⚠️ **TRAP:** "Minimize maintenance + needs Database Mail / Service Broker" → **SQL Managed Instance**, NOT SQL Database single/elastic pool. These features are NOT available in SQL Database.

🔗 **Azure Hybrid Benefit:** Existing SQL Server licenses with Software Assurance can reduce Azure SQL Database costs by up to 40%. This is the answer when asked about cost reduction with existing licenses.

📚 SQL Language Categories (DDL / DML / DCL)

Category	Full name	Statements
DDL	Data Definition Language	CREATE, ALTER, DROP, TRUNCATE, RENAME

DML	Data Manipulation Language	SELECT, INSERT, UPDATE, DELETE
DCL	Data Control Language	GRANT, REVOKE, DENY

⚠ **TRAP:** GRANT and REVOKE → **DCL**, NOT DML. DROP → **DDL**, NOT DML. INSERT → **DML** ✓

SQL Clause Order: SELECT → FROM → WHERE → GROUP BY → HAVING → ORDER BY

Memory: "Some Friends Watch Great Hollywood Operas"

🔒 ACID Properties

Letter	Property	Definition	Hook
A	Atomicity	All statements succeed or ALL roll back. No partial commits.	"Atom" = indivisible
C	Consistency	DB moves from one valid state to another. All rules/constraints respected.	"Consistent" = always valid
I	Isolation	Concurrent transactions don't see each other's uncommitted changes.	"Isolated" = separated
D	Durability	Once committed, data survives failures. Logged to disk.	"Durable" = permanent

⚠ **TRAP:** "All data changes will be logged and accounted for" → This is **Durability**, NOT Consistency. Consistency = DB stays in a valid state per its rules.

⚠ **TRAP:** "Processes can see only data in a consistent state" → This is **Isolation**, NOT Consistency.

🔑 Key SQL Objects

Object	Definition	Key fact
View	Virtual table defined by a saved SELECT query	No data stored; executes query on access
Stored Procedure	Saved SQL code block with logic (INSERT/UPDATE/DELETE allowed)	Reusable; can contain any SQL
Index	Data structure that speeds up lookups on a table	Like a book's index; improves SELECT performance
Heap	Table with NO clustered index	Data in no particular order
Clustered Index	Defines the physical order of rows in a table	Only 1 per table (it IS the table)

📐 Normalisation

Statement	True?
Normalisation eliminates relationships between tables	✗ NO — it defines and preserves them properly
Normalisation reduces data redundancy	✓ YES — splits into related tables
Normalisation improves data integrity	✓ YES — enforces consistent, accurate data

🔧 Query Tools

Tool	Works with
SSMS (SQL Server Management Studio)	SQL Server, Azure SQL DB, SQL Managed Instance, Synapse SQL pools

Azure Data Studio	SQL Server (incl. big data clusters), Azure SQL, Synapse, PostgreSQL (ext.)
MySQL Workbench	Azure Database for MySQL, Azure Database for MariaDB
pgAdmin	Azure Database for PostgreSQL

☞ **Rule:** Tools designed for an engine also work with Azure's managed version of that engine. MySQL Workbench works with MariaDB because MariaDB is a MySQL fork sharing the same protocol.

3 Non-Relational Data on Azure

~20%

🌐 Azure Cosmos DB APIs

API	Data model	Trigger word
NoSQL API (Core)	JSON documents	Default Cosmos DB, document store
MongoDB API	JSON documents (BSON)	MongoDB workloads, JSON + SQL queries
Gremlin API	Graph (nodes + edges)	"relationships between entities", networks, social graphs
Table API	Key/value pairs	"key/value", Azure Table Storage upgrade
Cassandra API	Column-family (wide-column)	High write throughput, Apache Cassandra workloads

⚠ **TRAP:** Key/value → **Table API**, NOT Cassandra API. Cassandra = wide-column (column-family). Very commonly confused!

☞ **Throughput levels in Cosmos DB:** Can be set at **database** level or **container** level. NOT at item or partition level.

📊 NoSQL Data Models

Model	Structure	Best for	Azure service
Document	Self-contained JSON/XML docs	Catalogs, user profiles, content	Cosmos DB (NoSQL/MongoDB API)
Key/Value	Unique key → value	Sessions, caching, preferences	Cosmos DB (Table API), Redis
Graph	Nodes + edges (relationships)	Social networks, fraud detection, network topology	Cosmos DB (Gremlin API)
Column-family	Rows with flexible columns	High write throughput, IoT, time-series	Cosmos DB (Cassandra API)

⚠ **TRAP:** "Relationships between utilisation levels of network devices" → **Graph** database. Not columnar! Graph = nodes + edges = network topology.

📁 Azure Storage Services

Service	Protocol/Access	Best for	Key fact
Blob Storage	REST / HTTP/HTTPS	Unstructured data (images, video, backups)	Object store; 3 tiers: Hot, Cool, Archive
Azure Files	SMB v3 / NFS	Cloud file shares, lift-and-shift file servers	Mount as network drive
Azure Table Storage	REST / OData	Key/value structured data	NoSQL; Cosmos DB Table API = premium version

Azure Queue Storage REST / HTTP Message queuing between services Decouples application components

⚠ **TRAP:** SMB protocol → **Azure Files**, NOT Blob Storage. Blob uses REST/HTTP. Azure Files is the only storage service mountable as a network drive via SMB.

🔗 **ADLS Gen2:** Azure Data Lake Storage Gen2 = Blob Storage + hierarchical namespace enabled. Used for big data analytics. Parquet files live here.

4 Analytics Workloads on Azure

~30%

⚙ Azure Analytics Services

Service	Purpose	Primary language	Key fact
Azure Synapse Analytics	Unified analytics platform (SQL + Spark)	SQL	Combines data warehouse + big data; outputs Parquet; serverless or dedicated SQL pools
Azure Data Factory (ADF)	Data integration & pipeline orchestration (ETL/ELT)	GUI / JSON	Move & transform data between services
Azure Databricks	Apache Spark analytics & ML platform	Python / Spark / Scala	Collaborative notebooks; ML workloads
Azure Stream Analytics	Real-time streaming data processing	SQL	Reads from Event Hubs / IoT Hub
Azure Data Explorer (ADX)	Real-time log & telemetry analytics	KQL	Time-series; also used in Azure Monitor, Sentinel
Azure HDInsight	Managed open-source clusters (Hadoop, Hive, Kafka)	HiveQL / Spark	Legacy big data workloads

⚠ **TRAP:** Azure Data Explorer uses **KQL** (Kusto Query Language) — NOT SQL. Stream Analytics and Synapse use SQL. ADX is the only major Azure analytics service using KQL.

🗨 Language → Service Mapping

Service	Primary Language	Remember because...
Azure Data Explorer	KQL	Unique Microsoft log/telemetry language
Azure Stream Analytics	SQL	SQL-like for real-time streams
Azure Synapse Analytics	SQL	Enterprise analytics = SQL core
Azure Databricks	Python / Spark	ML + developer workloads

🏗 Azure Data Factory — Activity Types

Activity type	Purpose	Examples
Data movement	Moves data from source to destination	Copy activity
Data transformation	Reshapes / processes data	Mapping data flow, Stored Procedure, Databricks Notebook
Control	Orchestrates pipeline logic & flow	Lookup , ForEach, If Condition, Wait, Execute Pipeline

⚠ **TRAP: Lookup = Control** activity — NOT transformation. It retrieves a value to drive pipeline logic. This is the most common ADF trap on the exam.

⚠ **TRAP: Mapping data flow = Data transformation** — NOT control. It reshapes data visually (joins, aggregations, filters).

★ Star Schema & Analytical Models

Concept	Definition
Fact table	Central table containing measurable data (sales amounts, quantities, revenue)
Dimension table	Surrounding tables with descriptive attributes (Date, Product, Customer, Geography)
Measure	Numeric value that can be aggregated (SUM, AVG, COUNT)
Dimension	Categorical attribute used to slice/filter data
Cube	Multidimensional structure formed where dimensions intersect
Star schema	Fact table surrounded by denormalised dimension tables
Snowflake schema	Like star schema but dimension tables are further normalised

⚠ **TRAP:** "Any point where dimensions intersect represents ___" → **a cube**, NOT an aggregated measure. The measure lives at the intersection; the cube is the structure itself.

🏠 Power BI Components

Component	Purpose
Power BI Desktop	Windows app to build reports and data models
Power BI Service	Cloud platform to publish, share, and collaborate
Power BI Mobile	iOS/Android app to view reports on the go
Power BI Report Builder	Tool for paginated reports (pixel-perfect, printable)
Paginated Report	Traditional-style report designed to be printed or exported to PDF
Dashboard	Single page of pinned visuals from multiple reports

🔗 **Dashboard content:** Can pin a report page, a single visualisation from a report, or a text box. A dashboard is a curated view — not a full report.

🔄 ETL vs ELT

	ETL	ELT
Order	Extract → Transform → Load	Extract → Load → Transform
Where transform	Outside the target (staging area)	Inside the target (data warehouse / lake)
Requirement	Schema defined before loading (schema on write)	Schema applied on read (schema on read)
Best for	Relational warehouses, sensitive data	Data lakes, big data, Synapse

⚠ **TRAP:** ETL requires "data that is fully processed BEFORE being loaded" — not a matching schema in source and target, and not that target must be relational.

🔍 High-frequency Gotchas

#	Question pattern	Wrong instinct	✓ Correct
1	Creating closed captions from audio	Descriptive	Cognitive analytics
2	Relationships between network devices	Columnar	Graph database
3	Key/value data in Cosmos DB	Cassandra API	Table API
4	ADF: Lookup activity type	Transformation	Control
5	ADF: Mapping data flow type	Control	Data transformation
6	ACID: "data changes logged and accounted"	Consistency	Durability
7	ACID: "processes see only consistent state"	Consistency	Isolation
8	Minimize maintenance + DB Mail + Service Broker	SQL Database elastic pool	SQL Managed Instance
9	SMB v3 protocol access	Blob Storage	Azure Files
10	Point where dimensions intersect	Aggregated measure	A cube
11	Azure Data Explorer language	SQL	KQL (Kusto Query Language)
12	PaaS: who applies OS patches?	You	Microsoft
13	Reduce Azure SQL cost with existing licenses	Not possible	Azure Hybrid Benefit
14	GRANT / REVOKE statements	DML	DCL
15	DROP statement	DML	DDL

🔍 Quick-Fire Reference Tables

Cosmos DB APIs → Data models

[Gremlin → Graph](#)
[MongoDB → JSON docs](#)
[Table → Key/value](#)
[Cassandra → Column-family](#)
[NoSQL → JSON docs](#)

ADF Activities

[Copy → Data movement](#)
[Mapping data flow → Transformation](#)
[Lookup → Control △](#)
[ForEach → Control](#)
[If Condition → Control](#)

SQL Categories

[SELECT → DML](#)
[INSERT → DML](#)
[UPDATE → DML](#)
[DELETE → DML](#)
[CREATE → DDL](#)
[DROP → DDL](#)
[ALTER → DDL](#)
[GRANT → DCL △](#)
[REVOKE → DCL △](#)

Analytics Languages

[ADX → KQL △](#)
[Stream Analytics → SQL](#)
[Synapse → SQL](#)
[Databricks → Python/Spark](#)

Azure Storage Protocols

[Blob → REST/HTTP](#)
[Azure Files → SMB v3 △](#)
[Table Storage → REST/OData](#)
[Queue → REST/HTTP](#)

